

Which Chinese Company Sold the Tunnel-Boring Machine to India?

Zhihu

First, as other Zhihu users have answered (Which Chinese company sold the tunnel-boring machine to India?), South Asia — that is, the great power of South Asia, India — also has tunnel-boring machine (TBM) plants set up by multinational corporations, and can manufacture TBMs domestically. Most recently, such a machine was used in the construction of the Mumbai Metro^[1]. China's tunnel-boring machines, though cheap, do not constitute a technological blockade. To compel Chinese firms not to sell would mean voluntarily ceding the market to other firms. Under the grand strategy of "dual circulation", intervening in the sale of products that carry no risk whatsoever of leaking advanced technology seems highly incongruous.

Second, the Indian enterprise BEML is developing the country's first domestically made TBM. This machine has a diameter of 6.5 metres and is intended mainly for metro construction. The project is still at the design stage, with the completion date unknown. The Delhi Metro Rail Corporation has already signed a memorandum of understanding with BEML for the purchase of TBMs, guaranteeing the machine a buyer. Both BEML and the Delhi Metro Rail Corporation are state-owned enterprises; it appears India is also preparing to replicate the Chinese model — mobilising the whole nation, as it were, to develop such a great instrument of national strength as the tunnel-boring machine.

Beyond TBMs, Indian enterprises have also made considerable progress in heavy equipment such as hydraulic excavators. In 2021, BEML unveiled a 180-tonne hydraulic excavator^{[2][3]}. Hydraulic excavators of the 200-tonne class were developed by Chinese firms in 2008. At present China's largest hydraulic excavator is a 700-tonne machine, while the world's largest hydraulic excavator is of the 980-tonne class, manufactured by Caterpillar.



BEML's largest bulldozer is of the 100-tonne class^[4], released in 2024^[5]. China's bulldozers of the same class were released around 2012.

So the momentum of India's mid-range industrial development is not to be underestimated.

Notes

[1] Thane–Borivali Twin Tunnel Project makes rapid progress, completion target set for 2028

中文原文：Thane–Borivali Twin Tunnel Project makes rapid progress, completion target set for 2028

<https://www.projectstoday.ai/News/ThaneBorivali-Twin-Tunnel-Project-makes-rapid-progress-completion-target-set-for-2028>

[2] Indian-made BEML 180 t class BE1800D hydraulic diesel driven shovel goes into production with Eastern Coalfields

中文原文：Indian-made BEML 180 t class BE1800D hydraulic diesel driven shovel goes into production with Eastern Coalfields

<https://im-mining.com/2021/06/29/indias-beml-180-t-class-be1800d-hydraulic-shovel-goes-production-eastern-coalfields/>

[3] SANY Group news page

<https://www.sanygroup.com/news/8954.html>

[4] BEML BD475-2-2 product page

<https://www.bemlindia.in/product/bd475-2-2/>

[5] BEML Ltd Unveils India's Biggest and Advanced Crawler Dozer – A Milestone in the Nation's Mining Growth
中文原文：BEML Ltd Unveils India's Biggest and Advanced Crawler Dozer – A Milestone in the Nation's Mining Growth
<https://www.bemlindia.in/wp-content/uploads/2024/12/pressrelese211224.pdf>

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